Nivedhitha Duggi

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Experience

Fixed Networks Customer Engineering Co-Op *Dallas, USA*
Nokia *Jan 2026 – Present*
Team:*NI S&D CD NAM&LAT CE FN*

- Designed and developed an intent-driven network and test automation framework for Nokia Altiplano within the Fixed Networks team, enabling automated incident detection and remediation by comparing expected versus real-time network configurations, reducing manual validation effort by 80% and improving observability across FTTH/GPON network infrastructure.
- Responsible for automation, validation, and deployment of AT&T MDU WiFi 7 service, integrating Nokia OLTs/ONTs with RGNets Gateway to deliver a cost-optimized, telco-grade reliable and secure solution
- Automated SDN cloud-controller installation and migration scripts, increasing speed by 1200% and eliminating errors drastically reducing downtime.
- Provided on-call support and incident management for production deployments, performing root-cause analysis and remediation for complex network issues using NETCONF/YANG and Layer 2/3 protocols.

Software Engineering Co-Op *Carrollton, USA*
Lennox International Inc *May 2025 – Dec 2025*
Team:*CQT team*

- Conducted S40 Thermostat Electronic Load Control, IFC board First Article Inspection, and flame-sensing threshold tests in collaboration with Controls-Quality and Test team, improving hardware validation accuracy by 35% across constant-torque and variable-speed HVAC systems.
- Analyzed 17,000+ returned thermostats for contamination and defect patterns using Python and Microsoft Excel automation, identifying root causes that reduced repeat field failures by 22%.
- Executed R2D2_ORCA refrigerant leak detection and Nydeck Perfect Speed motor compatibility tests, enhancing sensor calibration and safety diagnostics with 98% detection precision
- Migrated object detection models to edge devices for quality inspection, reducing bandwidth and infrastructure expenses by 15%. Developed AI-based scheduling for smart thermostats, reducing manual adjustments by 40%,and built an explainable agent and Qlik dashboard for model transparency and performance tracking.
- Upgraded the e-commerce portal by integrating Azure Cognitive Search and a GPT assistant, improving user experience and increasing conversions by 15%.Designed a Next.js-based AI lead management portal for HVAC dealers to prioritize leads, manage appointments, monitor missed calls & streamline communications.
- Enhanced an HR chatbot by implementing Azure AD authentication and secure API integration, improving knowledge retrieval speed and HR system efficiency. Collaborated with cross-functional teams to deliver scalable, production-ready solutions with LangChain, SQL, Python, and Microsoft Azure.

Python Developer Intern *Atlanta, USA*
Exponential AI *April 2024 – Aug 2024*
Team:*Solutions Team*

- Contributed to Machine Learning model advancements within the Solutions team for healthcare insurance products using ensemble techniques such as Random Forest and Deep Learning models, focusing on intelligent contract data extraction from healthcare documents.
- Succeeded in Applying cutting-edge models such as BERT, fine-tuned distil-BERT and RoBERTa for extractive Question Answering on scanned documents to extract sensitive keywords. Leveraged LLama cpp and Mistral for nuanced information retrieval, integrating NLTK, Spacy, and tuned Yolov8 for precise handwritten recognition.Carried out RAG and Agentic workflow for enhanced self data-driven decision-making capabilities, attaining increase in keyword extraction accuracy.
- Utilized VQAs like DONUT and LayoutLM with advanced quantized PEFT mechanisms integrated with

Layoutparser and Meta's Detectron2. Implemented multimodal img2table for extracting tabular data and leveraged TAPAS for querying, tailored to specific organizational needs. This approach improved data extraction speed through efficient code practices, enhancing document understanding and information retrieval capabilities in Provider Contracts documents.

Software Engineer Intern *Hyderabad, IN*
Practo *Jan 2024 – Apr 2024*
Team:*Product Engineering Team*

- Designed Clinical RAG system indexing 8,000 medical guidelines from S3 into Pinecone with specialty, care setting and recency filters, reducing clinician time-to-answer by 60% from 8.0 minutes 3.2 minutes).
- Cut LLM hallucinations by 72% through safety guardrails and rigorous A/B prompt testing, achieved 90% citation coverage on clinical queries.
- Implemented emotion and cognitive distortion detection using LangChain + LangGraph, enabling real-time personalization of CBT chat responses with PostgreSQL-backed user context storage.

Systems Engineer Intern *Chennai, IN*
Pragyashal Cloud Solutions *May 2023 – Nov 2023*
Team:*Founder's Office*

- Built secure,high-performance transaction interfaces using Angular 2+ and Redux, managing state for 10,000+ concurrent sessions with real-time balance updates in edtech environments.
- Refactored authentication workflows using JWT and Axios interceptors, reducing API failure rates by 15% through improved error handling and request management.
- Designed and implemented CI/CD pipelines using Jenkins and Docker, automating regression testing with Jest and Selenium to support reliable weekly releases.

Software Engineer Intern *Bengaluru, IN*
Team:*Marketing and Customer Engagement Team* *May 2022 – Aug 2022*
Pine Labs

- Developed and automated 15+ enterprise software solutions for Marketing and Customer Engagement team across 25 banking clients increasing customer feedback survey participation by 60%.
- Designed and implemented scalable automation workflows using Python, REST APIs, SFTP, and parallel processing, reducing manual effort by 80–90%, streamlining customer survey collection through WhatsApp integrations, and supporting platform growth from 1 million to 3 million merchants.
- Developed backend risk-monitoring and merchant management systems using behavioral analytics and user-tagging mechanisms, improving fraud detection accuracy and reducing fraudulent merchant activities by 95%.

Research & Publications

Graduate Student Research Assistant – Computational Healthcare and Biotechnology (COHB) Lab, UNT.
Denton, TX — Oct 2024 – May 2025
Conducted benchmark evaluations under Prof. Mohammed Aledhari using GLUE and SuperGLUE for healthcare-focused LLMs. Analyzed bias, fairness, and domain-specific performance metrics to enhance the ethical deployment and reliability of language models in clinical applications.

Developed and fine-tuned compact deep learning architecture for automated assay interpretation from smartphone images, achieving 94% accuracy in detecting SARS-CoV-2 and HCV with high sensitivity (4,000 and 2,200 copies/mL respectively).

Integrated AI with microfluidics and nanotechnology to create **VISTA** a disposable, electricity-free platform delivering lab-quality diagnostics in point-of-care and resource-limited settings in under 45 minutes.

Graduate Student Research Assistant – Department of Information Science, University of North Texas.
Denton, TX — Oct 2024 – May 2025
Collaborated with Prof. Sagnik Ray Choudhury to explore LLM-driven formative assessment tools using clinical datasets. Evaluated LLMs on MLMU and ARC for generating domain-specific MCQs and fill-in-the-blank questions. Applied statistical methods to assess relevance, contributing to innovations in medical education

technology.

Graduate Student Research Assistant – High Performance Cloud Computing (HPCC) Lab, University of North Texas. *Denton, TX — Oct 2024 – Jan 2025*
Benchmarked heterogeneous cloud compute services under Prof. Mohsen Amini Salehi, focusing on DNN inference, ML inference, and video transcoding. Collected and analyzed data on latency, cost, energy, and carbon footprint to optimize cloud resource selection.

Published findings in: **Benchmarking Different Application Types across Heterogeneous Cloud Compute Services**, arXiv, Jan 2025. This research contributed to optimizing performance and sustainability in cloud infrastructure for AI workloads.

Projects

Natural Language Processing with Disaster Tweets

- Developed a logistic regression model to classify disaster-related tweets using a dataset of 10,000 tweets. Preprocessed text data with tokenization, stemming, and stopword removal, then converted it into numerical features using TF-IDF. Tackled class imbalance with RandomOverSampler and evaluated results using precision, recall, and F1-score. Visualized model predictions and class distributions with Seaborn and Matplotlib, offering deeper insight into data trends. Achieved 80.31% accuracy and contributed to decision-making in disaster response scenarios
- **Technologies:** Python, Logistic Regression, NLTK, Scikit-learn, Pandas, Matplotlib.

Connect4 Reinforcement Learning

- Developed an intelligent agent for the Connect 4 game using Reinforcement Learning (RL) techniques, implementing both Q-Learning and Deep Q-Learning (DQL) algorithms to enable competitive gameplay against human and AI opponents. Leveraged OpenAI Gym to simulate game environments and efficiently manage complex state-action spaces. Utilized RL-based libraries such as Stable-Baselines3 for training, evaluation, and benchmarking agent performance. Focused on optimizing learning strategies to enhance the agent's decision-making and robustness during gameplay. The agent underwent extensive testing against a variety of AI opponents, consistently achieving a high win rate.
- **Technologies:** Python, Reinforcement Learning, Q-Learning, Deep Q-Learning, OpenAI Gym, Stable-Baselines3, Machine Learning

Recycrafter: AI-Powered Gamified Recycling Platform

- Developed an interactive 2D game using Python and Pygame, incorporating multi-level progression and real-time obstacle avoidance mechanics. Engineered adaptive gameplay dynamics, including collision detection, a scoring system, and difficulty scaling across three increasingly challenging levels. Designed custom visual assets, integrated background music, and implemented smooth game state transitions such as start, pause, and end screens. Optimized performance by reducing frame rendering latency by 25% and enhancing input handling to deliver more responsive and fluid player controls.
- **Technologies:** Python, Pygame, Git/GitHub, Object-Oriented Programming.

A Command-Line Interface for File Management

- Developed a command-line interface (CLI) program that emulates a Bash shell for efficient file and folder management, incorporating key functionalities such as add, move, alias, teleport, find, list, and quit. Engineered a left child–right sibling tree data structure to represent the directory hierarchy efficiently and implemented a hash table to manage directory aliases, enabling faster navigation and improved usability within the virtual file system. Designed a user-friendly menu interface with structured command prompts and clear feedback to enhance user interaction. Conducted thorough testing and debugging to ensure robust performance, reliability, and a smooth user experience.
- **Technologies:** C,Trees (Left Child-Right Sibling), Hash Table, GCC, Git

A twitter clone

- Developed Connect 2, a social media platform inspired by Twitter, allowing users to create, edit, delete, and share posts with real-time interaction. Designed and implemented secure user authentication features including account creation and login, ensuring data privacy and protection. Built a dynamic post management system to support seamless content engagement, with capabilities such as editing and deletion. Developed a responsive front-end to provide a consistent user experience across desktop and mobile devices. Integrated a robust back-end architecture using Node.js and Express.js, with MongoDB for real-time data storage and retrieval.

- **Technologies:** HTML, CSS, JavaScript, Node.js, Express.js, MongoDB, Git

Skills

Programming Languages: Python, C, C++, Java, JavaScript, HTML, CSS, SQL, R

Web Development: Django, React.js, Next.js, Node.js, Express.js, Flask, jQuery, Bootstrap

Machine Learning & Data Science: TensorFlow, Keras, PyTorch, Scikit-Learn, XGBoost, Pandas, NumPy, Jupyter Notebook, Anaconda, Matplotlib, Seaborn

NLP & LLMs: Hugging Face Transformers, LangChain, OpenAI APIs, SpaCy, NLTK, RAG Pipelines, Agentic AI Workflows, Prompt Engineering, LangGraph

Networking & Telecommunications : TCP/IP, SSL/TLS, HTTPS, IPSec, VPN, VLAN, QoS, NETCONF/YANG, SDN, Intent-Based Networking (IBN), Passive Optical Networks (PON), Nokia Altiplano, Nokia Lightspan OLTs, ONTs, Residential Gateways (RGs)

Cloud DevOps: Linux, Docker, Kubernetes, GCP(BigQuery), MongoDB, Kafka, AWS(S3,Glue,Athena,Lambda), MySQL, MongoDB, PostgreSQL,Snowflake, Azure(Data Factory, Synapse Analytics)

Tools & IDEs : Git, VS Code, Jira, Selenium, Postman, RESTful APIs, Tableau, PowerBI, MS Excel

Development Practices: Agile/Scrum, Software Development Lifecycle (SDLC), Test-Driven Development (TDD), CI/CD, Code Reviews, API Design, Cross-Functional Collaboration, Performance Optimization, System Debugging, Root Cause Analysis, Technical Documentation

Additional Skills : Data Structures & Algorithms, Image Processing, Test Automation, Unity, 3D Modelling, Blockchain(Ethereum, Solidity, Truffle, Web3.js, MetaMask)

Education

University of North Texas *Aug 2024 – May 2026*
MS in Computer and Information Science
Achievements: Distinguished Graduate Student for the 2024–2025 & 2025-2026 academic year
Coursework: Software Engineering,Software Development for AI,Computer Algorithms, Natural Language Processing, Machine Learning, Virtual Reality and Augumented Reality,Computer Networks,

Rajiv Gandhi University of Knowledge Technologies Nov 2020 – Apr 2024
Bachelors in Computer Science
Achievements: Dean's List 2022-2023, Dean's List 2023-2024
Coursework: Artificial Intelligence, Data Science with Python, Database Management System, Mathematical Foundation for DataScience, Operating Systems, SoftwareTesting